CHHAYA KUMARI

chhayachoudhary620@gmail.com | 9693160764 | linkedin.com/in/choudharychhaya | github.com/chhaya412

Summary

AI/ML Engineering student with hands-on experience building agentic and multimodal AI systems, including remote-sensing analysis platforms and AI-driven agricultural solutions. Skilled in Python, deep learning, NLP, and computer vision, with practical exposure to full-stack integration, cloud deployment, and prompt engineering. Seeking an internship/entry-level AI/ML Engineer role to apply machine learning and software engineering skills to real-world problems.

Technical Skills

- **Programming Languages:** Python, C++, SQL, NoSQL
- **AI/ML & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Agentic AI, Generative AI, Prompt Engineering
- **Tools & Platforms:** Git, GitHub, Docker, Jupyter Notebook, Google Colab, DevOps, Cloud Computing
- **Web & Backend:** React.js, Backend Development

Projects

GeoFusion AI — Agentic Multimodal Remote-Sensing Analysis Platform
Next.js, React, TypeScript, MUI, FastAPI, PostgreSQL, SQLAlchemy, Rasterio/GDAL, PyTorch, Hugging Face, BigEarthNet

- Designed and built an agentic multimodal AI platform that interprets natural-language queries to automatically select and run the appropriate remote-sensing analysis workflow.
- Implemented support for single-image, bi-temporal, and co-registered optical–SAR image analysis, enabling Visual Question Answering, scene description, and change detection.
- Applied vision-language and deep learning models (PyTorch, Hugging Face) adapted for remote-sensing data to generate evidence-grounded answers with visual outputs such as change maps and highlighted regions.
- Engineered a validation pipeline to check image format, metadata, spatial alignment, and large raster compatibility prior to model inference.
- Built GIS-compatible output generation, converting detected pixel masks into GeoJSON for downstream use in GIS software, along with confidence scoring and auditable execution traces.

AgriValue AI — AI-Driven Agricultural Value-Chain Platform
Next.js, React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, FastAPI, Scikit-learn, XGBoost, PyTorch, YOLO, ESP32, MQTT, Socket.IO, Docker, AWS

- Developed a full-stack platform connecting farmers directly with institutional buyers to reduce intermediaries and improve farmer income.
- Built AI-based demand and price forecasting models using Scikit-learn and XGBoost to support data-driven decision-making for farmers and buyers.
- Integrated IoT sensors (ESP32, temperature/humidity) with MQTT and Socket.IO for real-time cold-chain monitoring and logistics tracking.
- Implemented quality grading using deep learning (PyTorch, YOLO) and designed secure authentication (JWT, RBAC) and payment settlement flows.
- Architected cloud deployment using Docker, GitHub Actions, and AWS/Vercel for scalable, production-ready delivery.

Experience

Bharat Cares **6 Months**
Machine Learning Intern

- Worked on real-world machine learning projects, gaining practical experience in applying ML concepts and building industry-relevant technical skills.

Education

Rai University, Ahmedabad **CGPA: 8.75**
B.Tech in Computer Science and Engineering (AI/ML) *3rd Year*

Certifications & Achievements

- AI/ML Certification — Bharat Cares
- Participant, National-Level Hackathon — IIT Dhanbad
- Winner, State-Level Hackathon